

ClawScript Parser Expansion

Attila, here's the expanded parser code snippet in JavaScript (as the original was in JS for your Replit/Node setup). I've added:

- **Loops:** Full support for `LOOP num TIMES ... ENDLOOP` and `WHILE cond ... ENDWHILE` — parsed to `for/while` in JS.
- **Error Handling:** `TRY ... CATCH var ... ENDTRY` — parsed to `try-catch` blocks.
- **OpenClaw AI Tools Integration:** For commands like `AI_QUERY`, `ANALYZE_LOG`, `RUN_ML`, `CLAW_WEB`, etc., the generator adds imports (`require('../openclaw-ai.cjs')`) and generates `await` calls (e.g., `await ai.aiQuery(prompt, {tool, arg})`). Assume you have wrapper modules like `openclaw-ai.cjs` that export functions calling your OpenClaw tools (e.g., `aiQuery` uses `code_execution` or `conversation_search`).
- **General Improvements:** Better error messages with position/token info. Handled optional params (e.g., `TOOL`, `ARG`). Added more command parsers (stubs for many, full for key ones).

The parser is now more robust — test it in Replit by running

`parseAndGenerate(sample_code, 'TestStrategy')` to get the generated JS string, then write to file.

Expanded Code Snippet: `skills/bots/clawscript-parser.cjs`

```
JavaScript

const fs = require('fs');
const path = require('path');

// Token types
const TOKEN_TYPES = {
  KEYWORD: 'keyword',
  IDENTIFIER: 'identifier',
  STRING: 'string',
  NUMBER: 'number',
  OPERATOR: 'operator',
  PUNCTUATION: 'punctuation',
```

```

    COMMENT: 'comment'
};

// Lexer
function lexer(code) {
    const tokens = [];
    const regex = /\|\/\|.*|\/\|*[\s\S]*?*\|\/|"(?:\\\[^\]|\\^\\")*"|\d+\|.\|*\|([a-zA-Z_]\|w*| [= > < ! & | + - / * % ] + | [ ( ) { } [ \ ] ; , ] / g;
    let match;
    while ((match = regex.exec(code))) {
        let value = match[0];
        let type;
        if (value.startsWith('\/\|') || value.startsWith('\/*')) {
            type = TOKEN_TYPES.COMMENT;
        } else if (value.startsWith('"')) {
            type = TOKEN_TYPES.STRING;
            value = value.slice(1, -1);
        } else if (/^\d/.test(value)) {
            type = TOKEN_TYPES.NUMBER;
            value = parseFloat(value);
        } else if (/^[a-zA-Z_]/.test(value)) {
            type = TOKEN_TYPES.IDENTIFIER;
        } else if (/[= > < ! & | + - / * % ]/.test(value)) {
            type = TOKEN_TYPES.OPERATOR;
        } else {
            type = TOKEN_TYPES.PUNCTUATION;
        }
        if (type !== TOKEN_TYPES.COMMENT) {
            tokens.push({ type, value });
        }
    }
    return tokens;
}

// Parser
class ClawScriptParser {
    constructor(tokens) {
        this.tokens = tokens;
        this.pos = 0;
        this.ast = { type: 'Program', body: [] };
        this.variables = new Map();
        this.imports = new Set(['indicators']);
    }
}

```

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    }

    current() {
        return this.tokens[this.pos];
    }

    eat(type) {
        const token = this.current();
        if (token && token.type === type) {
            this.pos++;
            return token;
        }
        throw new Error(`Parse Error: Expected ${type}, got ${token ?
token.type : 'EOF'} at position ${this.pos}`);
    }

    peek() {
        return this.tokens[this.pos + 1];
    }

    parse() {
        while (this.pos < this.tokens.length) {
            const stmt = this.parseStatement();
            if (stmt) this.ast.body.push(stmt);
        }
        return this.ast;
    }

    parseStatement() {
        const token = this.current();
        if (!token) return null;

        const upperValue = token.value.toUpperCase();
        switch (upperValue) {
            case 'DEF':
            case 'SET':
                return this.parseVarDeclaration(upperValue === 'DEF');
            case 'BUY':
            case 'SELL':
            case 'SELLSHORT':
                return this.parseTradeCommand(upperValue);
            case 'EXIT':
            case 'CLOSE':

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        return this.parseExitCommand(upperValue);
    case 'TRAILSTOP':
        return this.parseTrailStop();
    case 'IF':
        return this.parseIfStatement();
    case 'LOOP':
    case 'WHILE':
        return this.parseLoop(upperValue);
    case 'AI_QUERY':
        return this.parseAIQuery();
    case 'AI_GENERATE_SCRIPT':
        return this.parseAIGenerate();
    case 'ANALYZE_LOG':
        return this.parseAnalyzeLog();
    case 'RUN_ML':
        return this.parseRunML();
    case 'CLAW_WEB':
        return this.parseClawWeb();
    case 'CLAW_X':
        return this.parseClawX();
    case 'CLAW_PDF':
        return this.parseClawPdf();
    case 'CLAW_IMAGE':
        return this.parseClawImage();
    case 'CLAW_VIDEO':
        return this.parseClawVideo();
    case 'CLAW_IMAGE_VIEW':
        return this.parseClawImageView();
    case 'SPAWN_AGENT':
        return this.parseSpawnAgent();
    case 'CALL_SESSION':
        return this.parseCallSession();
    case 'MUTATE_CONFIG':
        return this.parseMutateConfig();
    case 'ALERT':
        return this.parseAlert();
    case 'STORE_VAR':
        return this.parseStoreVar();
    case 'LOAD_VAR':
        return this.parseLoadVar();
    case 'WAIT':
        return this.parseWait();
    case 'TRY':
```

```

        return this.parseTryCatch();
    case 'ERROR':
        return this.parseError();
    case 'INCLUDE':
        return this.parseInclude();
    case 'OPTIMIZE':
        return this.parseOptimize();
    case 'INDICATOR':
        return this.parseIndicator();
    default:
        if (token.type === TOKEN_TYPES.IDENTIFIER) {
            return this.parseAssignment();
        }
        throw new Error(`Unexpected token: ${token.value}`);
    }
}

```

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parseVarDeclaration(isDef) {
    this.eat(TOKEN_TYPES.KEYWORD); // DEF/SET
    const name = this.eat(TOKEN_TYPES.IDENTIFIER).value;
    this.eat(TOKEN_TYPES.OPERATOR); // =
    const value = this.parseExpression();
    if (isDef && this.variables.has(name)) throw new Error(`Var ${name}
already defined`);
    this.variables.set(name, value);
    return { type: 'VarDecl', name, value, isDef };
}

```

```

parseTradeCommand(type) {
    this.eat(TOKEN_TYPES.KEYWORD); // BUY/SELL/SELLSHORT
    let size = 'config.size';
    if (this.current().type === TOKEN_TYPES.NUMBER) {
        size = this.eat(TOKEN_TYPES.NUMBER).value;
    }
    let orderType = 'MARKET';
    let price = null;
    if (this.current().value.toUpperCase() === 'AT') {
        this.eat(TOKEN_TYPES.KEYWORD); // AT
        orderType = this.eat(TOKEN_TYPES.IDENTIFIER).value.toUpperCase();
        if (orderType === 'LIMIT') {
            this.eat(TOKEN_TYPES.PUNCTUATION); // (
            price = this.parseExpression();
            this.eat(TOKEN_TYPES.PUNCTUATION); // )
        }
    }
}

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        this.eat(TOKEN_TYPES.FUNCTION); // /
    }
}
let condition = true;
if (this.current().value.toUpperCase() === 'IF') {
    this.eat(TOKEN_TYPES.KEYWORD);
    condition = this.parseExpression();
}
let stop = 'config.stopDistance';
if (this.current().value.toUpperCase() === 'STOP') {
    this.eat(TOKEN_TYPES.KEYWORD);
    stop = this.parseExpression();
}
let limit = 'config.limitDistance';
if (this.current().value.toUpperCase() === 'LIMIT') {
    this.eat(TOKEN_TYPES.KEYWORD);
    limit = this.parseExpression();
}
let reason = '"Auto"';
if (this.current().value.toUpperCase() === 'REASON') {
    this.eat(TOKEN_TYPES.KEYWORD);
    reason = this.parseExpression();
}
this.imports.add('trade');
return { type: 'Trade', command: type, size, orderType, price,
condition, stop, limit, reason };
}

parseLoop(loopType) {
    this.eat(TOKEN_TYPES.KEYWORD); // LOOP/WHILE
    let condition = null;
    if (loopType === 'WHILE') {
        condition = this.parseExpression();
    } else {
        const num = this.parseExpression();
        this.eat(TOKEN_TYPES.KEYWORD); // TIMES
        condition = { type: 'LoopCount', num };
    }
    const body = [];
    const endKey = loopType === 'LOOP' ? 'ENDLOOP' : 'ENDWHILE';
    while (this.current() && this.current().value.toUpperCase() !== endKey)
    {
        body.push(this.parseStatement());
    }
}

```

```

    {
    this.eat(TOKEN_TYPES.KEYWORD); // ENDLOOP/ENDWHILE
    return { type: 'Loop', loopType, condition, body };
    }

    parseTryCatch() {
    this.eat(TOKEN_TYPES.KEYWORD); // TRY
    const tryBody = [];
    while (this.current() && this.current().value.toUpperCase() !==
'CATCH') {
        tryBody.push(this.parseStatement());
    }
    this.eat(TOKEN_TYPES.KEYWORD); // CATCH
    const catchVar = this.eat(TOKEN_TYPES.IDENTIFIER).value;
    const catchBody = [];
    while (this.current() && this.current().value.toUpperCase() !==
'ENDTRY') {
        catchBody.push(this.parseStatement());
    }
    this.eat(TOKEN_TYPES.KEYWORD); // ENDTRY
    return { type: 'TryCatch', tryBody, catchVar, catchBody };
    }

    parseAIQuery() {
    this.eat(TOKEN_TYPES.KEYWORD); // AI_QUERY
    const prompt = this.eat(TOKEN_TYPES.STRING).value;
    let tool = null;
    if (this.current().value.toUpperCase() === 'TOOL') {
        this.eat(TOKEN_TYPES.KEYWORD);
        tool = this.parseExpression();
    }
    let arg = null;
    if (this.current().value.toUpperCase() === 'ARG') {
        this.eat(TOKEN_TYPES.KEYWORD);
        arg = this.parseExpression();
    }
    this.imports.add('ai');
    return { type: 'AIQuery', prompt, tool, arg };
    }

    // [Add similar parse methods for other commands, e.g., parseAlert,
    parseError, parseWait, parseClawWeb, etc. Stub example:]
    parseAlert() {

```

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    this.eat(TOKEN_TYPES.KEYWORD); // ALERT
    const message = this.parseExpression();
    let level = 'info';
    if (this.current().value.toUpperCase() === 'LEVEL') {
        this.eat(TOKEN_TYPES.KEYWORD);
        level = this.parseExpression();
    }
    let to = null;
    if (this.current().value.toUpperCase() === 'TO') {
        this.eat(TOKEN_TYPES.KEYWORD);
        to = this.parseExpression();
    }
    this.imports.add('alert');
    return { type: 'Alert', message, level, to };
}

// ... (implement the rest as per list, using similar patterns)

generateJS(ast, strategyName) {
    let js = `const BaseStrategy = require('./base-strategy.cjs');\n`;
    this.imports.forEach(imp => js += `const { ${imp} } = require('./` +
openclaw-${imp}.cjs`);
    js += `\n` + `class ${strategyName}Strategy extends BaseStrategy {\n`;
    js += `  constructor(config) { super(config); }\n\n`;
    js += `  async evaluateEntry(ticks, context) {\n`;
    ast.body.forEach(stmt => js += this.generateStmt(stmt, '    '));
    js += `  return null;\n  }\n\n`;
    js += `  getRequiredBufferSize() { return 50; }\n`;
    js += `  getConfigSchema() { return []; }\n`;
    js += `  static get STRATEGY_TYPE() { return "custom-` +
`${strategyName.toLowerCase()}`; }\n`;
    js += `};\n\nmodule.exports = ${strategyName}Strategy;\n`;
    return js;
}

// [Expanded generateStmt, generateExpr as before]

}

// Main function
function parseAndGenerate(code, strategyName) {
    const tokens = lexer(code);
    const parser = new ClawScriptParser(tokens);

```



```

const parser = new StrategyParser({engine});
const ast = parser.parse();
return parser.generateJS(ast, strategyName);
}

// Example test (run in Replit)
const sample = `
DEF risk = 2

LOOP 3 TIMES

    TRY

        DEF rsi_val = INDICATOR(RSI, 14)

        IF rsi_val < 30 THEN

            BUY 1 AT MARKET STOP 10 LIMIT 20 REASON "Low RSI"

        ENDIF

        AI_QUERY "Analyze" TOOL "tool" ARG "arg"

    CATCH err

        ALERT "Error" LEVEL "error" TO "telegram"

        ERROR "Failed"

    ENDTRY

    WAIT 1000

ENDLOOP

WHILE risk > 0

    SET risk = risk - 1

ENDWHILE
`;

const generated = parseAndGenerate(sample, 'FullTest');

```

```
console.log(generated);  
</parameter>  
</xai:function_call>
```